

Running the Chess Engine as a Lichess Bot on ZimaOS

The engine repo (github.com/michiklein/chess-engine) ships with a Docker setup that runs the engine behind the lichess-bot bridge. One container is the whole bot: start the container and the bot is online on Lichess, stop it and it goes offline. Resource caps are built in (1.5 CPU cores, 512 MB RAM), so it will not crowd out other services on the server.

1. One-time: Lichess account and token (in a browser)

a) Create a FRESH Lichess account at lichess.org - bot accounts must never have played a human game. Example name: MichaelsEngineBot.

b) Logged in as that account, create an API token at the address below. The bot:play scope is pre-selected. Copy the token - it is shown only once.

```
https://lichess.org/account/oauth/token/create
?scopes[]=bot:play&description=lichess-bot
```

c) Upgrade the account to a BOT (one-time, irreversible for that account). From any machine:

```
curl -d '' https://lichess.org/api/bot/account/upgrade \
-H "Authorization: Bearer YOUR_TOKEN"
```

2. One-time: deploy on the ZimaOS server

Open a terminal on the server (SSH or the ZimaOS web terminal) and run:

```
git clone https://github.com/michiklein/chess-engine.git
cd chess-engine
LICHESS_BOT_TOKEN=YOUR_TOKEN docker compose up -d --build
```

The first build takes a few minutes - it compiles the engine from source inside the container. Check that it is alive with:

```
docker logs -f chess-lichess-bot          # Ctrl-C stops watching
```

You should see the lichess-bot banner and a line saying the bot is connected.

3. Daily use: taking the bot on/off

- The container appears in the ZimaOS dashboard as chess-lichess-bot with a start/stop toggle. Stop = bot offline on Lichess. Start = back online.

- After a server reboot it starts again automatically, unless it was deliberately stopped.

4. Playing against it

- Anyone can open the bot's Lichess profile and click "Challenge to a game". It accepts standard chess, casual and rated, bullet through classical, one game at a time.

- To make it auto-challenge other bots when idle, edit `docker/config.yml` in the repo (matchmaking: `allow_matchmaking: true`) and redeploy as in section 5.

5. Updating the bot after engine improvements

```
cd chess-engine
git pull
```

```
LICHESS_BOT_TOKEN=YOUR_TOKEN docker compose up -d --build
```

Avoid updating mid-game (the game would be abandoned) - check the bot's Lichess profile for a running game first.

6. Troubleshooting and notes

- Container exits immediately: run `docker logs chess-lichess-bot`. The usual cause is a missing or wrong token - the container requires the `LICHESS_BOT_TOKEN` environment variable.
- "Too Many Requests" in the logs: harmless rate limiting; the bridge retries by itself.
- The token is a secret: anyone who has it can play as the bot. It lives only in the shell command/environment, never in git.